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Full Stack Python Developer

SUMMARY

I am a Full Stack Python Developer with almost 7 years of experience in the IT field, specializing in designing, developing, and deploying scalable applications. I bring strong analytical programming skills and a deep understanding of various tools, frameworks, and technologies. My experience spans web and application development, cloud computing, data analytics, and containerization. My dedication to delivering top-notch software solutions and knowledge of software development principles make me a valuable asset to any team or project.

- **Python and Django Proficiency:** Extensive expertise in Python, particularly with Django and Flask frameworks. Proficient in creating web applications using the Model-View-Controller (MVC) architecture, building scalable APIs, and designing interactive web solutions.
- **Front-End Development:** Skilled in crafting responsive and visually engaging user interfaces using technologies such as HTML5, CSS3, JavaScript, Typescript, Bootstrap, React.js, Angular.js, and jQuery.
- **Database Management:** Adept at working with relational databases such as MySQL and Oracle, including writing complex SQL queries, PL/SQL functions, stored procedures, and optimizing database performance.
- **Cloud Computing:** Hands-on experience with cloud platforms like AWS (EC2, S3, Lambda, CloudFront) and Microsoft Azure (Azure Functions, App Services, Logic Apps, Blob Storage). Proficient in automating cloud workflows and implementing CI/CD pipelines using Azure DevOps, Jenkins, and Terraform.
- **Big Data and Analytics:** Proficient in Big Data frameworks, including Hadoop and Spark (Spark Core, Spark SQL, DataFrames, and PySpark). Experienced in data analysis with Pandas and visualization using Tableau.
- **Containerization and Orchestration:** Expertise in Docker, Kubernetes, and Red Hat OpenShift for containerized application deployment and management. Skilled in creating and maintaining containerization scripts and configurations for scalability.
- **Object-Oriented Programming:** Deep understanding of OOP principles, multithreading, and ORM libraries for effective database interaction.
- **Version Control and Collaboration:** Proficient in version control using GIT and experienced in Agile methodologies, collaborating in sprint-based development.
- **Tools and Frameworks:** Experience with Python libraries such as NumPy, PyQt, Matplotlib, and Flask. Familiar with tools like JIRA, Confluence, Wireshark, and Fiddler2 for development and debugging.
- **Web Services:** Developed and utilized web services using REST and SOAP protocols, ensuring seamless integration between applications.
- **Linux and Windows Environments:** Proficient in scripting and development on both Linux (Debian, Ubuntu, Red Hat) and Windows systems.
- **Capital Market:** Good understanding of various capital markets products such as equity, options, mutual funds, fixed income, annuities, new issue management, currency, next shares for retail as well as managed clientsegies.

WORK EXPERIENCE

Python Developer
Bank Of America, Pennington, NJ

AUG 2024 – PRESENT

Responsibilities:

- Developed Python scripts to parse and transform XML data into structured formats for email processing and compliance.
- Extracted and processed structured data from XML using `xml.etree.ElementTree`, handling encoding issues and surrogate Unicode characters.
- Developed RESTful APIs using FastAPI to expose XML parsing and email processing functionality for integration with internal tools and services.
- Designed authentication and authorization mechanisms for the API using OAuth2 and API key strategies to protect sensitive email data.
- Implemented email domain mapping logic to classify and route messages based on internal and external recipients.
- Automated attachment processing, encoding and embedding files using Base64 for secure inclusion in email and archival formats.
- Worked with MIME-based email formatting, ensuring proper structuring of email files for compatibility with enterprise email systems.
- Designed structured logging mechanisms for tracking errors, debugging XML transformations, and monitoring data processing pipelines.
- Optimized XML parsing performance using efficient Python data structures and regular expressions for data extraction.
- Implemented message grouping logic, consolidating records based on unique identifiers to maintain structured conversation history.
- Handled event-based message processing, distinguishing between user messages and system-generated events within XML records.
- Developed recipient classification logic, ensuring correct assignment of email attributes and metadata modifications for compliance.
- Validated and enriched XML schema transformations, ensuring output met industry standards and business requirements.
- Used Bitbucket for version control, maintaining structured commit histories for Python scripts.
- Tracked and managed development tasks in Jira, including XML transformation enhancements, bug fixes, and automation improvements.

Environment: Python, XML, RESTful APIs, FastAPI, EML, MIME, Regular Expressions (Regex), Bitbucket, JIRA, Git, Linux, Bash, JSON, Logging, Unicode Handling, Agile, Data Processing, Automation, File Handling.

Full Stack Python Developer
American Express, New York, NY

OCT 2022 – MAY 2024

Responsibilities:

- Developed and consumed APIs using Python's requests module in Django and Flask for JSON RESTful services and Murex data integration.
- Built and modified Python/Django modules, database models, APIs, and views for interactive web solutions using Django Database APIs.
- Designed and developed web pages using HTML, CSS, JavaScript, TypeScript, React.js, Angular.js, Node.js, Vue.js, jQuery, Ajax, and Bootstrap.
- Created React.js applications with JSX, virtual DOM, routers, and dynamic forms, ensuring UI compliance with HTML5, CSS3, and JavaScript standards.
- Built scalable distributed data solutions using Hadoop, Spark Core, Spark SQL, PySpark, DataFrames, and Pandas for big data processing.
- Automated AWS services (EC2, S3, DynamoDB, ELB, CloudFront) configuration and deployed Flask/Django microservices on AWS (EBS and EC2).
- Automated tasks using Bash and Python scripts, including creating a generic stored-procedure-as-a-service microservice.
- Managed AWS resources (EC2, S3, Lambda) and developed RESTful APIs deployed on cloud platforms.
- Developed containerized applications using Docker, Kubernetes, and Red Hat OpenShift for scalable deployment and orchestration.
- Used Python Pandas for data manipulation and Tableau for creating dashboards and analyzing datasets through custom SQL queries.

- Automated Murex trade validation and reporting processes with Python for enhanced accuracy and efficiency.
- Conducted Oracle performance tuning using AWR and Explain Plans and wrote optimized SQL queries for complex data processing.
- Transitioned applications from Windows to Linux servers and employed Kafka for real-time data streaming and messaging.
- Debugged, troubleshoot, and optimized web applications using tools like Wireshark, live HTTP headers, and Fiddler2.
- Designed GUIs using webapp2 to dynamically showcase documentation and features via web browsers.
- Developed scalable microservices in Flask, Django, and Spring Boot for cloud-based environments.
- Managed team collaboration using GIT, JIRA, and Confluence for version control, issue tracking, and user management.
- Created and managed containerization configurations and CI/CD platforms, supporting deployment, scaling, and system backups.
- Analyzed and processed financial data for asset management, risk analytics, and economic capital using emerging market strategies.

Environment: Python, Django, Flask, FastAPI, AWS (EC2, S3, CloudFront, EBS, DynamoDB, Lambda), Murex, Hadoop, Spark (Spark Core, Spark SQL, DataFrames, Spark Streaming), PySpark, Pandas, HTML, CSS, TypeScript, JavaScript (React.js, Angular.js, Vue.js, Node.js), jQuery, Bootstrap, Tableau, Jenkins, Terraform, Docker, Kubernetes, Red Hat OpenShift, Spring Boot, Oracle DB, MySQL, JSON, Kafka, Big Data, Microservices, JIRA, Confluence, webapp2, Wireshark, Fiddler2, Linux, Git

Python Developer

MAY 2020 - AUG 2022

Goldman Sachs, New York, NY

Responsibilities:

- Developed Python-based risk analytics tools and trade data pipelines.
- Built automated workflows for financial data processing and pricing models, optimizing scalability.
- Created web applications using the Django Framework, adhering to the MVC architecture, and managed databases with Django Database API and MySQL.
- Designed horizontally scalable APIs and handled CRUD operations using Django-REST Framework and ViewSets.
- Designed cloud workflows using Microsoft Azure Functions, App Services, Logic Apps, and managed Azure resources like Blob Storage and SQL Databases.
- Implemented CI/CD pipelines in Azure DevOps, Jenkins, and Terraform, automating deployments and ensuring compliance with best practices.
- Designed and developed microservices-based applications using Spring Boot and Spring Integration frameworks.
- Utilized PyQt and PyGUI for interactive user interface development, including creating and modifying client reports.
- Built dynamic web pages using React.js, Angular.js, jQuery, and integrated UI designs with HTML5, CSS3, JavaScript, TypeScript, Bootstrap, and JSON.
- Used React.js to develop components with JSX for data binding, forms, and routers, enabling enhanced interactivity.
- Operated within Agile environments, collaborating with cross-functional teams and managing JIRA and Confluence tools.
- Implemented a test-driven development approach, creating unit tests using Python's unittest framework.
- Developed Python and UNIX shell scripts for automation, establishing job processes and performing MySQL queries.
- Conducted debugging and troubleshooting across Windows and Linux environments for web applications.
- Optimized database performance with MySQL and Oracle, leveraging SQL tuning techniques and integrating libavg for advanced UI development.

Environment: Python, Django, Flask, Microsoft Azure (Azure Functions, Azure App Services, Logic Apps, Azure DevOps, Azure SQL Database, Blob Storage, Virtual Machines), PyQt, PyGUI, libavg, JIRA, JIRA Service Desk, Confluence, Spring Boot, Microservices, React.js, Angular.js, Agile, HTML5, CSS3, JavaScript, TypeScript, Bootstrap, jQuery, JSON, Django REST Framework, Bash/UNIX Shell Scripting, RESTful APIs, MySQL, Oracle DB, Linux, Windows, GitHub.

Python Developer

OCT 2018 - APR 2020

Citizens Bank, San Francisco, CA

Responsibilities:

- Developed web applications using the Django framework with MVC architecture, and implemented scalable APIs using Django-REST framework.
- Created Python and UNIX shell scripts for automation, debugging, and data processing.
- Managed databases using Django database API, MySQL, and Oracle, including creating complex objects like Tables, Views, and Sequences.

- Designed dynamic user interfaces with HTML5, CSS3, JavaScript, TypeScript, Bootstrap, Angular.js, and jQuery.
- Automated repetitive tasks and workflows using Amazon EC2 command-line interface, Bash, and Python scripts.
- Utilized containerization technologies like Docker, Kubernetes, and Red Hat OpenShift for deployment, orchestration, and scalability.
- Enhanced application performance and ensured container security, networking, and orchestration.
- Developed multi-threaded Python applications and used PyQt for interactive UI features and reporting.
- Integrated data between MS Excel and Oracle tables and utilized ORM libraries for efficient database interaction.
- Executed MySQL queries via Python and performed data visualization tasks.
- Loaded and processed large datasets using MySQL and Oracle Forms and Reports.
- Employed test-driven development practices and debugged web applications across multiple environments.
- Worked in an Agile development environment, contributing to the complete project lifecycle from design to deployment.
- Designed and implemented scripts for monitoring and enhancing circuit parameters and hardware environments.
- Utilized JSON for data exchange and integrated RESTful APIs for seamless CRUD operations.

Environment: Environment: Python, Django, Flask, MySQL, Oracle DB, HTML5, CSS3, JavaScript, TypeScript, Bootstrap, JSON, jQuery, Angular.js, RESTful API, SOAP, PyQt, UNIX/Linux, GIT, Kubernetes, Docker, Red Hat OpenShift, Amazon EC2, Bash, ORM libraries, and Agile.

EDUCATION

Masters in Data Science | Saint Peter's University | Jersey City, New Jersey, USA **SEPT 2022 - MAY 2024**

Bachelors in Technology | ABES Engineering College | Ghaziabad, Uttar Pradesh, India **AUG 2014 - SEPT 2018**